

FOURIER TRANSFORM IN DIGITAL IMAGE PROCESSING

1. The Core Idea: From Spatial Domain to Frequency Domain

Spatial Domain: This is what we normally see. An image is represented by pixels, where each pixel has a location (x, y) and an intensity value. We process the image by directly manipulating these pixel values (e.g., adjusting contrast, applying filters).

Frequency Domain: The Fourier Transform allows us to represent the image in terms of its spatial frequencies.

What is a Spatial Frequency? It describes how quickly pixel values change across the image.

Low Frequencies: Correspond to slow-varying, smooth areas of the image (e.g., a cloudless sky, a blank wall). They define the overall structure and energy of the image.

High Frequencies: Correspond to rapidly changing areas, such as edges, fine textures, noise, and sharp details.

The Fourier Transform is a mathematical tool that decomposes an image from its spatial domain into its constituent sine and cosine waves of different frequencies, directions, and amplitudes.

Fourier Transform Formula :

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

. Forward Fourier Transform:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

It breaks the image into its sinusoidal frequency components.

. Inverse (Reverse) Fourier Transform:

Converts frequency domain → spatial domain, Used to reconstruct the image from its frequency representation.

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Spatial Domain: Pixel intensity (direct image)

Frequency Domain : Rate of change of intensity.